

DBMS – SUPPLIER AND INVENTORY MANAGEMENT

Project Title: Vilvah – Natural Skincare and Personal Care Database Management System

Introduction

The Vilvah Supplier and Inventory Management System is designed to manage supplier information and product inventory for the Vilvah natural skincare and personal care database.

The system maintains important supplier details such as Supplier ID, Supplier Name, Contact Person, Phone, Email, City and GST Number. It also manages inventory information including Inventory ID, Product ID, Quantity, Minimum Stock, Warehouse and Last Updated.

The system establishes relationships between **Suppliers, Products and Inventory** to maintain accurate product and stock information. It also helps track available and unavailable products, update inventory quantities and generate inventory status reports.

This module improves inventory management by providing clear information about product availability, stock levels, suppliers and warehouse details.

1. Database Design

1.1 Supplier Table

Attribute	Data Type	Key	Description
Supplier_ID	INT	PK	Unique supplier ID
Supplier_Name	VARCHAR(100)	-	Supplier name
Contact_Person	VARCHAR(100)	-	Contact person
Phone	VARCHAR(15)	-	Supplier phone number
Email	VARCHAR(100)	-	Supplier email
City	VARCHAR(50)	-	Supplier city
GST_Number	VARCHAR(20)	-	GST number

1.2 Inventory Table

Attribute	Data Type	Key	Description
Inventory_ID	INT	PK	Unique inventory ID
Product_ID	INT	FK	References Product
Quantity	INT	-	Available quantity
Minimum_Stock	INT	-	Minimum required stock
Warehouse	VARCHAR(100)	-	Warehouse location
Last_Updated	DATE	-	Last updated date

2 . Table Creation

2.1 Create Supplier Table

```
SQL> CREATE TABLE Supplier (  
2     Supplier_ID INT PRIMARY KEY,  
3     Supplier_Name VARCHAR(100) NOT NULL,  
4     Email VARCHAR(100) NOT NULL UNIQUE,  
5     Contact_No VARCHAR(15) NOT NULL UNIQUE,  
6     Address VARCHAR(200) NOT NULL  
7 );  
  
Table created.
```

2.2 Create Inventory Table

```
SQL> CREATE TABLE Inventory (  
 2     Inventory_ID NUMBER(10) PRIMARY KEY,  
 3     Product_ID NUMBER(10) NOT NULL,  
 4     Supplier_ID NUMBER(10) NOT NULL,  
 5     Stock_Quantity NUMBER(10) NOT NULL,  
 6     Stock_Status VARCHAR2(20) NOT NULL,  
 7     Last_Updated DATE NOT NULL,  
 8     FOREIGN KEY (Product_ID) REFERENCES Product(Product_ID),  
 9     FOREIGN KEY (Supplier_ID) REFERENCES Supplier(Supplier_ID),  
10     CHECK (Stock_Quantity >= 0),  
11     CHECK (Stock_Status IN ('Available', 'Unavailable'))  
12 );  
  
Table created.
```

2.3 Supplier Data Insertion

```
SQL> INSERT INTO Supplier  
 2 (Supplier_ID, Supplier_Name, Email, Contact_No, Address)  
 3 VALUES  
 4 (201, 'Green Earth Supplies', 'greeneearth@gmail.com', '9876543210', 'Chennai');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO Supplier  
 2 (Supplier_ID, Supplier_Name, Email, Contact_No, Address)  
 3 VALUES  
 4 (202, 'Natural Care Products', 'naturalcare@gmail.com', '9876543211', 'Coimbatore');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO Supplier  
 2 (Supplier_ID, Supplier_Name, Email, Contact_No, Address)  
 3 VALUES  
 4 (203, 'Organic Life Suppliers', 'organiclife@gmail.com', '9876543212', 'Madurai');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO Supplier  
 2 (Supplier_ID, Supplier_Name, Email, Contact_No, Address)  
 3 VALUES  
 4 (204, 'Pure Nature Supplies', 'purenature@gmail.com', '9876543213', 'Salem');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO Supplier  
 2 (Supplier_ID, Supplier_Name, Email, Contact_No, Address)  
 3 VALUES  
 4 (205, 'Eco Fresh Products', 'ecofresh@gmail.com', '9876543214', 'Trichy');  
  
1 row created.
```

```

SQL> INSERT INTO Inventory
  2 VALUES (301, 1001, 201, 50, 'Available', SYSDATE);
1 row created.

SQL>
SQL> INSERT INTO Inventory
  2 VALUES (302, 1002, 202, 40, 'Available', SYSDATE);
1 row created.

SQL>
SQL> INSERT INTO Inventory
  2 VALUES (303, 1003, 203, 30, 'Available', SYSDATE);
1 row created.

SQL>
SQL> INSERT INTO Inventory
  2 VALUES (304, 1004, 204, 0, 'Unavailable', SYSDATE);
1 row created.

SQL>
SQL> INSERT INTO Inventory
  2 VALUES (305, 1005, 205, 25, 'Available', SYSDATE);
1 row created.

SQL> SELECT * FROM Inventory;

```

2.4 Inventory Data Insertion

```

SQL> INSERT INTO Inventory
  2 VALUES (301, 1001, 201, 50, 'Available', SYSDATE);
1 row created.

SQL>
SQL> INSERT INTO Inventory
  2 VALUES (302, 1002, 202, 40, 'Available', SYSDATE);
1 row created.

SQL>
SQL> INSERT INTO Inventory
  2 VALUES (303, 1003, 203, 30, 'Available', SYSDATE);
1 row created.

SQL>
SQL> INSERT INTO Inventory
  2 VALUES (304, 1004, 204, 0, 'Unavailable', SYSDATE);
1 row created.

SQL>
SQL> INSERT INTO Inventory
  2 VALUES (305, 1005, 205, 25, 'Available', SYSDATE);
1 row created.

```

```

SQL> SELECT * FROM Inventory;

```

INVENTORY_ID	PRODUCT_ID	SUPPLIER_ID	STOCK_QUANTITY	STOCK_STATUS
301	1001	201	50	Available
302	1002	202	40	Available
303	1003	203	30	Available
304	1004	204	0	Unavailable
305	1005	205	25	Available

2.5 Supplier Product Information

```
SQL> SELECT
2      s.Supplier_ID,
3      s.Supplier_Name,
4      p.Product_ID,
5      p.Product_Name,
6      i.Stock_Quantity,
7      i.Stock_Status,
8      i.Last_Updated
9  FROM Supplier s
10 JOIN Inventory i
11   ON s.Supplier_ID = i.Supplier_ID
12 JOIN Product p
13   ON i.Product_ID = p.Product_ID;
```

SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
DATE	1991	
DATE	1991	beginning
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
VEHICLE COUNTRY	US	17-020-28
	30 available	
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
MAJOR	2000	Products
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
North	1000	Available
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
Argentine	1000	beginning
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
RAVE	1000	beginning
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
PLANNED	1000	beginning
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,196	
AVI	2000	Products
	30 available	17-020-28
SAPLN_03_10		
SAPLN_03_0905		
PRODUCT_10		
PRODUCT_0905		
STOCK_QUANTITY STOCK_030105	1,047,	

3. Inventory Status

3.1 Display Available Products

```
SQL> SELECT
2      p.Product_ID,
3      p.Product_Name,
4      i.Stock_Quantity,
5      i.Stock_Status
6 FROM Product p
7 JOIN Inventory i
8      ON p.Product_ID = i.Product_ID
9 WHERE i.Stock_Status = 'Available';
```

```

PRODUCT_ID
-----
PRODUCT_NAME
-----
STOCK_QUANTITY STOCK_STATUS
-----
      1001
Vitamin C Face Serum
      50 Available

      1002
Matte Lipstick
      40 Available

PRODUCT_ID
-----
PRODUCT_NAME
-----
STOCK_QUANTITY STOCK_STATUS
-----

      1003
Anti Dandruff Shampoo
      30 Available

      1005
Body Lotion

PRODUCT_ID
-----
PRODUCT_NAME
-----
STOCK_QUANTITY STOCK_STATUS
-----

      25 Available

```

4. Display Unavailable Products

```

SQL> SELECT
  2   p.Product_ID,
  3   p.Product_Name,
  4   i.Stock_Quantity,
  5   i.Stock_Status
  6 FROM Product p
  7 JOIN Inventory i
  8   ON p.Product_ID = i.Product_ID
  9  WHERE i.Stock_Status = 'Unavailable';

PRODUCT_ID
-----
PRODUCT_NAME
-----
STOCK_QUANTITY STOCK_STATUS
-----
      1004
Floral Eau De Parfum
           0 Unavailable

```

5. Inventory Update

5.1 Update Inventory Quantity

```

SQL> UPDATE Inventory
  2  SET Stock_Quantity = 60,
  3      Stock_Status = 'Available',
  4      Last_Updated = SYSDATE
  5  WHERE Inventory_ID = 301;

1 row updated.

```

6. Inventory Status Reports

6.1 Complete Inventory Report

```
SQL> SELECT
2     i.Inventory_ID,
3     p.Product_Name,
4     s.Supplier_Name,
5     i.Stock_Quantity,
6     i.Stock_Status,
7     i.Last_Updated
8 FROM Inventory i
9 JOIN Product p
10    ON i.Product_ID = p.Product_ID
11 JOIN Supplier s
12    ON i.Supplier_ID = s.Supplier_ID
13 ORDER BY i.Inventory_ID;
```

6.2 Available and Unavailable Product Count

INVENTORY_ID			
PRODUCT_NAME			
SUPPLIER_NAME			
STOCK_QUANTITY	STOCK_STATUS		LAST_UPDA
381			
Vitamin C Face Serum			
Green Earth Supplies	68 Available		17-SEP-26
INVENTORY_ID			
PRODUCT_NAME			
SUPPLIER_NAME			
STOCK_QUANTITY	STOCK_STATUS		LAST_UPDA
382			
Nutte Lipstick			
Natural Care Products	48 Available		17-SEP-26
INVENTORY_ID			
PRODUCT_NAME			
SUPPLIER_NAME			
STOCK_QUANTITY	STOCK_STATUS		LAST_UPDA
383			
Anti Dandruff Shampoo			
Organic Life Suppliers	38 Available		17-SEP-26
INVENTORY_ID			
PRODUCT_NAME			
SUPPLIER_NAME			
STOCK_QUANTITY	STOCK_STATUS		LAST_UPDA
384			
Floral Eau De Parfum			
Pure Nature Supplies	8 Unavailable		17-SEP-26
INVENTORY_ID			
PRODUCT_NAME			
SUPPLIER_NAME			
STOCK_QUANTITY	STOCK_STATUS		LAST_UPDA
385			
Body Lotion			
Eco Fresh Products	25 Available		17-SEP-26

6.3 Total Stock by Supplier

```
SQL> SELECT
2     Stock_Status,
3     COUNT(*) AS Product_Count
4 FROM Inventory
5 GROUP BY Stock_Status;
```

6.4 Products with Zero Stock

STOCK_STATUS	PRODUCT_COUNT
Available	4
Unavailable	1

7. Total Stock by Supplier

```
SQL> SELECT
  2     s.Supplier_ID,
  3     s.Supplier_Name,
  4     SUM(i.Stock_Quantity) AS Total_Stock
  5 FROM Supplier s
  6 JOIN Inventory i
  7   ON s.Supplier_ID = i.Supplier_ID
  8 GROUP BY
  9     s.Supplier_ID,
 10     s.Supplier_Name
 11 ORDER BY s.Supplier_ID;
```

```
SUPPLIER_ID
-----
SUPPLIER_NAME
-----
TOTAL_STOCK
-----
      201
Green Earth Supplies
      60

      202
Natural Care Products
      40

SUPPLIER_ID
-----
SUPPLIER_NAME
-----
TOTAL_STOCK
-----
      203
Organic Life Suppliers
      30

      204
Pure Nature Supplies

SUPPLIER_ID
-----
SUPPLIER_NAME
-----
TOTAL_STOCK
-----
      0

      205
Eco Fresh Products
      25
```

8. Products with Zero Stock

```
SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     s.Supplier_Name,
  5     i.Stock_Quantity,
  6     i.Stock_Status
  7 FROM Inventory i
  8 JOIN Product p
  9   ON i.Product_ID = p.Product_ID
 10 JOIN Supplier s
 11   ON i.Supplier_ID = s.Supplier_ID
 12 WHERE i.Stock_Quantity = 0;
```

```
PRODUCT_ID
-----
PRODUCT_NAME
-----
SUPPLIER_NAME
-----
STOCK_QUANTITY STOCK_STATUS
-----
      1004
Floral Eau De Parfum
Pure Nature Supplies
      0 Unavailable
```

10.Conclusion

The Vilvah Supplier and Inventory Management System successfully manages supplier and inventory information and establishes relationships between suppliers, products and stock.

Primary key and foreign key constraints help maintain data integrity and connect the Supplier, Product and Inventory tables. The system supports supplier data insertion, inventory management, stock updates and identification of available and unavailable products.

The inventory status reports provide useful information about product quantities, stock levels, warehouses and suppliers. Overall, this module provides a structured approach to managing supplier and inventory information in the Vilvah DBMS project.